

BinderDex

In-Depth Technical Specification

Pokémon TCG Collection Management & Creative Binder Platform

Version 4.0 | December 2025

Attribute	Details
Status	Awaiting Approval
Development Model	Solo developer + Claude Code
Timeline	21 weeks (Phase 0-7)
Primary Stack	Next.js 14, Supabase, TypeScript, Magic UI, Stripe
Key Change in v4	Unified Binder Builder (ChromaDex + Michi Mode)

1. Project Overview & Vision

1.1 Executive Summary

BinderDex is a web application for Pokémon TCG collectors to organize, track, and visualize their collections. The platform differentiates itself through the Unified Binder Builder, which combines algorithmic page generation (ChromaDex) with full creative control (Michi Mode).

1.2 Core Value Pillars

Pillar	Description
Browse	Comprehensive card database with powerful multi-filter search
Track	Visual master set tracker showing physical binder layout
Build	Unified builder with manual creation + algorithmic generation
Share	Community gallery, templates, and social export options

2. Technical Architecture

2.1 Technology Stack

Layer	Technology	Rationale
Framework	Next.js 14	App Router, Server Components, optimal Vercel deployment
Language	TypeScript	Type safety, better DX, runtime error reduction
Styling	Tailwind + Magic UI	Utility-first CSS, pre-built animated components
State	Zustand + React Query	Client state + server state caching
Database	Supabase PostgreSQL	Full SQL, RLS, real-time, 500MB free tier
Auth	Supabase Auth	Built-in JWT, OAuth providers, guest mode
Storage	Supabase Storage	User uploads, generated exports, 1GB free
Hosting	Vercel	Edge network, automatic HTTPS, preview deployments
Payments	Stripe	Subscriptions, webhooks, customer portal
Drag & Drop	@dnd-kit/core	Modern, accessible, touch-friendly
Image Processing	Cropper.js + node-vibrant	Client-side cropping, color extraction

2.2 Key Architectural Decisions

Decision 1: Unified Binder Builder Architecture

Context: ChromaDex (algorithmic) and Michi Method (manual) serve different needs but share infrastructure.

Decision: ONE shared canvas editor with two entry points. ChromaDex generates into the same editor as manual creation.

Implications: Users can blend approaches, algorithm output is always editable, reduced code duplication.

Decision 2: Extended Slot Model for Michi Features

Context: Michi Method requires slot merging and custom image uploads.

Decision: Add content_type, span_cols, span_rows, custom_image_url, crop_data to binder_slots.

Implications: Flexible slot content, backward-compatible, enables full-art spreads.

3. Database Schema

3.1 Core Tables

sets

```
id, name, series, era, printed_total, total, release_date,
has_reverse_holos, logo_url, symbol_url
```

cards

```
id, set_id, name, number, rarity, supertype, subtypes[],
types[], hp, artist, national_dex_numbers[], image_small,
image_large, is_promo, is_premium, isLegendary, isMythical, generation
```

card_variants

```
id, card_id, variant_type (NORMAL/REVERSE_HOLO/FIRST_EDITION), image_url
```

3.2 ChromaDex Tables

card_colors

```
id, card_id (UNIQUE), dominant_hex, dominant_hsl (JSONB),
palette (JSONB), brightness, saturation, warmth
```

3.3 Binder Tables (Updated for Unified Builder)

binders

```
id, user_id, name, description, slot_config,
type (CUSTOM/MASTER_SET/CHROMADEX/MICHI),
master_set_id, is_public, is_curated, template_id
```

binder_slots (UPDATED)

```
id, page_id, position (0-8/0-11/0-15),
content_type (CARD/CUSTOM_IMAGE/EMPTY/MERGED), -- NEW
variant_id (nullable),
custom_image_url TEXT, -- NEW
span_cols INTEGER DEFAULT 1, -- NEW
span_rows INTEGER DEFAULT 1, -- NEW
crop_data JSONB -- NEW: {x, y, zoom, rotation}
```

binder_templates (NEW)

```
id, name, description, category, slot_config,
layout_data (JSONB), is_premium, preview_image_url, created_by
```

custom_images (NEW)

```
id, user_id, storage_path, original_filename, file_size, uploaded_at
```

4. Unified Binder Builder Specification

4.1 Architecture Overview

The Unified Binder Builder provides two entry points (ChromaDex and Michi Mode) that both output to a shared canvas editor. Users can start with either approach and seamlessly blend algorithmic suggestions with manual creativity.

4.2 Shared Canvas Editor

- **Left Panel - Card Picker:** Search, filters, scrollable card grid, upload button
- **Right Panel - Binder Canvas:** Visual grid, droppable slots, page nav, ChromaDex button

4.3 Michi Mode Features

Slot Merging:

1. User multi-selects adjacent slots
2. Click Merge button
3. Anchor slot gets span_cols/span_rows > 1
4. Covered slots marked as MERGED

Custom Image Upload:

1. Click Upload Image button
2. Select PNG/JPG (max 5MB)
3. Image uploads to Supabase Storage
4. Cropper.js modal for positioning
5. Drag image to slot, stores URL + crop_data

4.4 ChromaDex Generation

Color Extraction Pipeline:

- Batch job during data import using node-vibrant
- Extract dominant + palette → Convert to HSL → Store in card_colors
- Scoring: CIEDE2000 color distance + diversity bonus

Generation Modes:

Mode	Description
Color - Dominant	Match cards with similar dominant color
Color - Palette	Match cards within a color gradient
Color - Harmony	Color wheel rules: complementary, analogous, triadic

In-Canvas Options:

- Generate for entire page
- Generate for empty slots only
- Generate for selected slots

5. Development Phases

Phase	Weeks	Deliverables
0	1-2	Foundation: Next.js, Supabase, Auth, CI/CD, Data pipeline
1	3-4	Card Browser: Grid view, shared filters, search, card modal
2	5-7	Master Set Tracker: Visual binder, preferences, progress, variants
3	8-11	Unified Builder Core: Canvas editor, card picker, drag-drop, Michi features
4	12-14	ChromaDex Integration: Color extraction, generation algorithm, in-canvas UI
5	15-16	Export/Import: CSV/PNG, PDF generation, TCGPlayer/Collectr parsers
6	17-19	Community: Gallery, sharing, templates library, social export
7	20-21	Monetization & Launch: Stripe, feature gating, ads, launch prep

6. Monetization

Feature	Guest	Free	Pro \$4.99
Browse Cards	✓	✓	✓
Master Set Trackers	1	1	Unlimited
Custom Binders	1 (10pg)	1 (10pg)	Unlimited
Slot Merging	✗	✓	✓
Custom Image Upload	✗	5 total	Unlimited
ChromaDex Generation	✗	3/month	Unlimited
Templates	Basic	Basic	All
Basic Export (CSV/PNG)	✗	✓	✓
Premium Export (PDF)	✗	✗	✓

7. Approval Checklist

- Supabase as sole backend
- Unified Binder Builder (ChromaDex + Michi Mode)
- Extended binder_slots schema
- 21-week timeline
- Initial sets: Mega Evolution Base + Phantasmal Flames
- Data source: PokemonTCG/pokemon-tcg-data GitHub repo

--- End of Technical Specification ---